# MATRIX First Release Manual

Operational guide for the implemented MATRIX tools

MATRIX Development Notes

June 28, 2026

**Abstract**

This manual defines the first release boundary of `MATRIX` and gives the operational contract for the implemented tools. `MATRIX` is the framework: each tool reads and writes shared enriched `xyzin` sections, uses shared libraries for repeated tasks, and can run independently when its required sections are already present. Fragment search, nano-LEGO assembly and the production TRINITY optimizer are reserved for the next release; this release is closed by the preprocessing, GIC, GF/PED, MORPHEUS, rovibrational, anharmonic, DVR, QM-adapter and GUI workflows documented here.

## Contents

# 1 Release Boundary

The first `MATRIX` release is a refactored, documented and testable replacement for the corresponding implemented parts of Merlino/ORACLE. The release is complete when the following conditions hold:

- every implemented scientific tool can run from the command line using only its declared `xyzin` sections;
- the GUI launches the same public commands and does not contain private parsers or scientific algorithms;
- all external file formats enter through one adapter per format;
- the same libraries are used for geometry, topology, symmetry, normal-mode data, diagonalization and publication exports;
- manuals identify consumed sections, produced sections, standalone commands, GUI placement and regression tests;
- CLI help, manual links and GUI help panels are generated from the same tool-contract metadata used by the `xyzin` readiness checks.

The next release owns robust fragment search, nano-LEGO fragmentation and assembly, LCB25 fragment indexing, the production TRINITY external energy/gradient optimization loop, PySCF launch/analysis adapters, VSCF for local anharmonic oscillators, vibronic spectroscopy, ionization potentials, electron affinities and expanded help content for new tools. The current release keeps the fragment and TRINITY schemas, command skeletons, GUI entry points and online help for implemented tools so downstream design does not need another reorganization.

# 2 Shared Container Contract

The central object is the enriched `xyzin` file. Each tool appends or updates sections instead of passing private scratch formats between modules. A tool must not reparse Gaussian, Molpro, ORCA, MRCC, SMILES or geometry data if the corresponding shared section is already present.

| Section | Owner | Main consumers |
|---|---|---|
| `#BASIC` | LINK / chemistry layer | all molecular tools |
| `#SYMMETRY` | LINK | NEO/GICForge, GUI |
| `#TOPOLOGY` | LINK | synthons, fragments, GICs |
| `#SYNTHONS` | LINK | GF/PED, MORPHEUS, future fragments |
| `#GIC`, `#SYCART` | NEO/GICForge | GF/PED, MORPHEUS, TRINITY |
| `#CARTESIAN_HESSIAN` | QM adapters | GF/PED, diagnostics |
| `#NORMAL_MODES` | QM adapters | vibrational comparison, hybrid spectra |
| `#QFF` | QM adapters | VPT2/VCI |
| `#ROTATIONAL` | rovib/QM adapters | WMS-Rot, thermo |

| | | |
|---|---|---|
| #VIBRATIONAL | rovib/QM adapters | spectra, thermo, DOS |
| #GF_PED | GF/PED | GUI, vibrational workbench |
| #MORPHEUS | MORPHEUS/SEFit | GUI, reports |
| #THERMO | thermo | publication exports |
| #VPT2_VCI | VPT2/VCI | vibrational workbench |
| #DVR | DVR | GUI, post-run analysis |

The practical rule is simple: if two tools need the same information, that information belongs in a shared section and in a shared library, not in two tool-local parsers.

## 3 LINK and Preprocessing

LINK, formerly ORACLE-Babel, is the preprocessing gate. It obtains an initial Cartesian geometry by reading an XYZ file directly, importing from a supported QM format, or building from SMILES through RDKit. It then performs symmetry detection, topology analysis, synthon typing and validation once, and writes the results back to the same `xyzin` container.

```
matrix link preprocess input.xyz molecule.xyzin
matrix link preprocess smiles.txt molecule.xyzin --source-kind auto
oracle validate molecule.xyzin
```

The output sections are normally `#SOURCE`, `#BASIC`, `#SYMMETRY`, `#TOPOLOGY` and `#SYNTHONS`. Downstream tools must consume those sections rather than re-detecting bonds, rings, point groups or atom classes.

Avogadro remains the preferred manual 3D inspection and editing tool. After the user accepts the molecular structure, the GUI displays the 2D molecular view and all later tasks work from the enriched `xyzin` state.

## 4 QM Adapters

QM adapters are the only code allowed to understand external program outputs. Gaussian, Molpro, ORCA and MRCC data are promoted into shared sections. Scientific tools consume these sections and never scan log files privately.

```
matrix gaussian promote-fchk calc.fchk molecule.xyzin
matrix gaussian promote-rovib calc.log molecule.xyzin
matrix gaussian promote-electronic calc.log molecule.xyzin
matrix molpro promote molpro.out molecule.xyzin
matrix orca promote orca.out molecule.xyzin
matrix mrcc promote mrcc.out molecule.xyzin
```

The main promoted sections are `#CARTESIAN_HESSIAN`, `#NORMAL_MODES`, `#QFF`, `#ROTATIONAL`, `#VIBRATIONAL`, `#DELTABVIB`, `#ELECTRONIC`, `#TRANSITIONS` and `#ORBITALS`. Orbital files are passed to Molden, Avogadro or MOrbVis-compatible viewers through the GUI command layer.

General QM properties are stored in `#PROPERTIES`. This section is owned by `matrix-qm` and records the property name, target, atom or fragment selector, isotope, values, units, axes, source program, method, level and any conversion label. It is meant for quantities whose availability and units depend on the external code, for example raw Molpro electric-field gradients and converted nuclear quadrupole coupling constants. Downstream spectroscopic tools must read this normalized section instead of reparsing QM outputs.

### 4.1 Remote QM Execution

Heavy QM jobs can be launched on the private Linux host `oracle` and then copied back for adapter promotion. The operational details are in `oracle_qm_remote_manual.pdf`. The current remote contract is:

- use only `gdv32` or `g16` for Gaussian jobs;
- load Molpro as `module load molpro/2025.3`;
- load ORCA through the configured ORCA module;
- submit long jobs through `matrix-submit` when possible;
- copy completed logs, FCHK files and outputs back to the local MATRIX workspace;
- promote those files through `matrix gaussian ...`, `matrix molpro ...` or the corresponding QM adapter before any scientific tool consumes the data.

The remote machine is an execution backend, not a second parser stack. The single-adapter-per-format rule is unchanged.

## 5 NEO / GICForge

The implemented GIC tool is still exposed as `gicforge` in the command line, with `neo` as an alias. The planned tool name is NEO, "Nonredundant Equivariant Orthogonalizer". The specialist manual `gicforge_manual.pdf` remains the detailed reference for primitive construction, ring puckering, symmetry projectors, special fragment-aware coordinates, B-matrix construction and Python/Fortran parity.

```
matrix gicforge molecule.xyzin
matrix neo molecule.xyzin
oracle gic-bmatrix oracle.gic.definition.json geometry.xyz
```

NEO consumes `#VALIDATION`, `#SYMMETRY`, `#TOPOLOGY`, `#SYNTHONS` and optional `#FRAGMENTS`. It produces `#GIC` and `#SYCART`. Symmetry-adapted GIC labels begin with the irrep label; totally symmetric blocks can therefore be selected directly in optimization or least-squares workflows. Symmetrization is a typed projector: stretches, bends, torsions, out-of-plane coordinates, ring-puckering blocks and protected special coordinates are never mixed just because they share an irrep. For ring-puckering source spaces, the projector is also checked against the analytic B-row space used downstream, so the frozen labels, Gaussian export and Wilson B matrix remain the same mathematical object.

## 6 GF/PED

GF/PED consumes a frozen GIC basis and a Cartesian Hessian. The specialist manual `gf_manual.pdf` is the detailed reference for Wilson GF theory, internal force constants, PEDs, local force-field approximations and Pulay-style scaling of force constants in GIC space.

```
matrix gf --xyzin molecule.xyzin --out gf.report
matrix gf --xyzin molecule.xyzin --csv-dir gf_tables
```

The module writes #GF_PED. It can subtract electrostatic and van der Waals non-bonded terms before a local force-field model is built. Frequency scaling is not a MATRIX vibrational-spectrum operation; scaling belongs here, at the force-constant level, where diagonal class scaling and geometric-mean off-diagonal scaling are chemically meaningful. When symmetry-block solution is requested, GF/PED first verifies that the internal $F$ and $G$ matrices are block diagonal in the stored GIC irreps. Couplings between different irreps are treated as a contract error in the coordinate/symmetry layer, not as a recoverable GF fallback.

# 7 MORPHEUS / SEFit

MORPHEUS performs semiexperimental geometry refinement. It includes the SEFit single-structure workflow and the multi-structure class-correction workflows. The detailed manuals are morpheus_manual.pdf and multistructure_manual.pdf.

```
matrix semiexp --xyzin molecule.xyzin --job job.toml --outdir run
matrix semiexp-ensemble ensemble.toml --outdir ensemble_run
matrix semiexp-benchmark snapshot.json --outdir benchmark
```

MORPHEUS consumes isotopologue data and optional #GIC, #SYCART or #MORPHEUS state. It writes #MORPHEUS, diagnostics, CSV tables and paper-oriented summaries. The local se_geometries library is an implemented support library; large-scale fragment search and nano-LEGO assembly remain next-release work.

# 8 Rotational Spectroscopy / WMS-Rot

The rotational workbench uses the local WMS-Rot Hamiltonian engine imported from the group web service. MATRIX does not reimplement the Hamiltonian. It normalizes constants, dipoles and vibration-rotation corrections through #ROTATIONAL, #VIBRATIONAL and #DELTABVIB, then calls the local engine.

```
matrix rovib summarize molecule.xyzin
matrix rovib wmsrot-input molecule.xyzin --out wmsrot.in
matrix rovib wmsrot-run molecule.xyzin --out rotational_lines.csv
```

The run writes line lists suitable for plotting and publication export. Future database comparison should query microwave/submillimeter catalogs through a shared experimental-spectrum adapter, not from GUI-local code. The planned catalog targets are CDMS, JPL and Splatalogue, with molecule identifiers, frequency windows, isotopic species and provenance saved in export metadata.

# 9 Vibrational Spectroscopy

The vibrational workbench plots IR, Raman, VCD and ROA spectra when the corresponding harmonic or anharmonic data are present in #VIBRATIONAL. It can compare spectra from one or two xyzin files. For ordinary IR and Raman comparisons, the second trace is plotted with negative intensity. VCD and ROA are signed observables and are therefore not mirrored.

```
matrix rovib vib-spectrum molecule.xyzin --observable IR \
  --source harmonic --csv ir.csv --plot ir.svg

matrix rovib vib-compare level1.xyzin level2.xyzin \
```

```
  --observable IR --first-source harmonic --second-source anharmonic \
  --csv compare.csv --plot compare.svg
```

Hybrid spectra are implemented as

$$\nu_{\mathrm{hybrid}} = \nu_{\mathrm{harm}}(\mathrm{level1}) + [\nu_{\mathrm{anh}}(\mathrm{level2}) - \nu_{\mathrm{harm}}(\mathrm{level2})] \, .$$

Before applying the correction, MATRIX reads `#NORMAL_MODES` from both files and performs a global assignment that maximizes the absolute normal-mode overlap. If any assigned overlap is below the threshold, the hybrid spectrum is rejected.

```
matrix rovib vib-spectrum level1.xyzin --source hybrid \
  --level2-xyzin level2.xyzin --observable IR \
  --csv hybrid.csv --plot hybrid.svg \
  --mode-match-csv mode_matches.csv --min-mode-overlap 0.70
```

Experimental NIST IR spectra are fetched automatically only when the NIST JCAMP record declares a gas-phase state. Condensed-phase records, missing records and unparseable records are returned to the GUI as states requiring user instructions.

```
matrix rovib nist-ir 74-82-8 --out methane_nist_ir.csv
```

## 10 Thermochemistry and Kinetics

The implemented thermochemistry tool consumes normalized molecular, rotational and vibrational sections and writes `#THERMO`. The GUI Thermo/Kinetics workbench displays the generated thermochemical tables and rovibrational density-of-states outputs. Production kinetics models and `#KINETICS` are a future extension boundary, not a blocker for this release.

```
matrix thermo molecule.xyzin --out thermo.report
matrix rovib dos molecule.xyzin --out dos_vib.dat
matrix rovib dos-rovib molecule.xyzin --out dos_rovib.dat --q-out rovib_qt.dat
```

Publication exports should keep temperature, pressure, symmetry number, frequency set, rotational constants and data provenance in the table metadata.

## 11 VPT2 / VCI

The VPT2/VCI package consumes `#QFF`. Gaussian FCHK or QFF text is only an adapter input; standalone operation reads the normalized `xyzin` section. Reports and CSV tables are attached to the vibrational workbench.

```
matrix vpt2-vci --xyzin molecule.xyzin --run-dir vpt2_run
matrix vpt2-vci --xyzin molecule.xyzin --run-dir vci_run --model vci
```

VPT2/VCI uses the shared diagonalization policy where matrix diagonalization is the dominant cost. GPU acceleration is delegated to the shared `oracle_core.diagonalizer` layer when supported by the local machine.

## 12 DVR

DVR workflows prepare and run scan/path DVR calculations and then normalize the post-run outputs into the GUI state. Large Hamiltonian diagonalizations are routed through the same shared diagonalizer used by other MATRIX modules.

```
matrix dvr prepare molecule.xyzin --scan-log scan.log --run-dir dvr_run
matrix dvr run --xyzin molecule.xyzin --run-dir dvr_run
matrix dvr status molecule.xyzin
```

The implemented GUI state is `#DVR`. New DVR engines must not add private diagonalization wrappers; they should call the shared diagonalizer so CPU/GPU policy is controlled in one place.

## 13 ORACLE GUI

The GUI keeps the ORACLE name in the MATRIX framework. Its planned expansion is "Operator for Routing, Analysis, Control, Launch and Exploration". The GUI owns project state, command construction, missing-section guidance, file selection, external viewer launches and publication export orchestration. It does not own scientific parsers or algorithms.

```
python -m matrix_oracle molecule.xyzin
matrix-oracle molecule.xyzin
```

The first-release GUI exposes:

- project dashboard and section readiness;
- structure/synthon/topology inspection and Avogadro handoff;
- NEO/GICForge, GF/PED, MORPHEUS, VPT2/VCI, DVR and TRINITY panels;
- rotational, vibrational, electronic and thermochemistry workbenches;
- orbital viewer launch through Molden, Avogadro or MOrbVis-compatible external viewers;
- missing-section suggestions that point to the owning adapter or tool.

## 14 Regression and Release Tests

The release gate is the full test suite plus focused smoke tests for workflows that cross tool boundaries. The minimum command is:

```
source scripts/matrix_env.sh
matrix-set
python -m ruff check packages tests
python -m pytest -q
```

Important cross-module checks include:

- Python/Fortran NEO/GICForge primitive, GIC and B-row agreement;
- Gaussian ReadAllGIC closed-loop checks using final log geometry, promoted Cartesian Hessian, strict GF symmetry blocks and PED irrep purity;

- GF/PED writing and reading `#GF_PED`;
- MORPHEUS standalone and multi-structure runs;
- WMS-Rot local line-list generation;
- vibrational mirror comparison and hybrid level1+level2 normal-mode matching;
- VPT2/VCI and DVR diagonalizer routing;
- GUI command builders matching the public CLI.

## 15 First Release Closure

With this manual and the specialist manuals present, the first `MATRIX` release can be declared closed when the repository is clean, the full tests pass, and the manuals are generated from source. The next release then starts from a clear scope: robust fragments, nano-LEGO, LCB25 fragment indexing, production TRINITY optimization, PySCF adapters, VSCF, vibronic spectroscopy, IP/EA workflows and expanded help/manual coverage for those new tools.